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EXPERIMENT 2

Aim: To extract key linguistic components, such as noun phrases, verb phrases, and named entities, from text using both traditional chunking methods and advanced NLP techniques.

Theory:

1. Natural Language Processing (NLP)

Natural Language Processing (NLP) is a field of artificial intelligence (AI) focused on the interaction between computers and human language. It involves teaching machines to understand, interpret, and generate human language in a way that is valuable and meaningful. Some common tasks in NLP include:

- **Tokenization:** Breaking down text into words or tokens.
- **Part-of-Speech (POS) Tagging:** Assigning grammatical categories (noun, verb, adjective, etc.) to each word.
- **Named Entity Recognition (NER):** Identifying entities (such as people, locations, dates, etc.) in text.
- **Dependency Parsing:** Analyzing grammatical structures and the relationships between words in a sentence.

2. Phrase Extraction

Phrase extraction involves identifying and extracting meaningful segments or "chunks" of text that serve a specific syntactic role. In NLP, these chunks can be **noun phrases (NP)**, **verb phrases (VP)**, **adjective phrases (AdjP)**, **adverb phrases (AdvP)**, **prepositional phrases (PP)**, and more.

Noun Phrases (NP):

- A **noun phrase** is a syntactic unit centered around a **noun** and its modifiers (e.g., adjectives, determiners, etc.).
- Example: In the sentence "*The quick brown fox*", the noun phrase is "*The quick brown fox*" because it consists of the noun *fox* and the adjectives *quick* and *brown*, along with the article *the*.

Verb Phrases (VP):

- A **verb phrase** typically includes the main verb and auxiliary verbs (helping verbs) that modify the action.
- Example: In the sentence "*She is running quickly*", the verb phrase is "*is running*", where *is* is the auxiliary verb and *running* is the main verb.

Adjective Phrases (AdjP):



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- An **adjective phrase** consists of an adjective and any modifiers that provide more details about the noun it is modifying.
- Example: In "*The incredibly fast runner*", the adjective phrase is "*incredibly fast*".

Adverb Phrases (AdvP):

- An **adverb phrase** is made up of an adverb and its modifiers.
- Example: In "*She runs quite quickly*", the adverb phrase is "*quite quickly*".

Prepositional Phrases (PP):

- A **prepositional phrase** consists of a preposition and its object, which can be a noun phrase.
- Example: In "*She sat on the chair*", the prepositional phrase is "*on the chair*".

3. Named Entity Recognition (NER)

Named Entity Recognition (NER) is a subtask of information extraction that seeks to locate and classify named entities in text into predefined categories, such as:

- **Person names:** *Barack Obama, John Smith*
- **Geopolitical entities (GPE):** *New York City, USA*
- **Organizations (ORG):** *Google, Microsoft*
- **Dates and times:** *March 1st, 2025, yesterday*
- **Monetary values:** *\$50, €30*

NER helps systems understand the real-world context of a text, making it essential for information extraction, question answering, and document classification.

4. Chunking in NLP

Chunking refers to dividing text into smaller, meaningful units, typically at the level of syntactic structures (e.g., noun chunks, verb chunks). **Chunking** involves grouping words that are grammatically related to each other, based on **POS tagging** or **dependency parsing**.

Chunking Methods:

- **Regular Expressions (Regex):** A simple way to identify patterns in text. For example, using regex to detect noun phrases might look for patterns such as capitalized words followed by adjectives and nouns.
- **Rule-based Chunking:** Uses predefined grammar rules to recognize and extract specific syntactic units.
- **Statistical Chunking:** Involves training models on labeled data to learn the chunking rules.

5. SpaCy (Advanced NLP Method)



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SpaCy is a powerful and efficient open-source NLP library in Python, built to handle a variety of NLP tasks, such as tokenization, part-of-speech tagging, named entity recognition, dependency parsing, and more.

SpaCy's main features that help in phrase extraction include:

- **Part-of-Speech Tagging:** Identifying the grammatical categories of words (e.g., noun, verb, adjective, etc.).
- **Dependency Parsing:** Understanding the syntactic structure of a sentence and how words relate to each other. This helps identify **verb phrases**, **noun phrases**, and relationships between words.
- **Named Entity Recognition (NER):** Automatically recognizing named entities in text.
- **Noun Chunks:** **SpaCy** has a built-in mechanism to extract noun phrases directly using the `.noun_chunks` attribute.
- **Pre-trained Models:** **SpaCy** provides pre-trained models for multiple languages, trained on large datasets for high accuracy.

6. Comparing Chunking and NLP (SpaCy)

- **Chunking** (using regex or rule-based systems) is **faster** and simpler but is **limited** to predefined patterns and often lacks the ability to capture complex relationships between words. It is typically used for simpler tasks, such as detecting specific phrase types in structured text.
- **SpaCy** (NLP-based method) uses **machine learning** models and **deep learning** techniques, making it much more **robust and flexible** for understanding complex sentences and extracting accurate syntactic structures like noun phrases, verb phrases, and named entities. It's also better at **scaling** for larger datasets and more diverse linguistic features.

Use Cases for Phrase Extraction:

- **Information Retrieval:** Extracting key phrases and named entities to improve search and retrieval systems.
- **Text Summarization:** Identifying important phrases (such as noun and verb phrases) for generating concise summaries.
- **Chatbots and Virtual Assistants:** Extracting named entities and other relevant phrases to understand user queries.
- **Sentiment Analysis:** Identifying sentiment-bearing phrases and entities to gauge public opinion or customer sentiment.



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Code:

1)

```
import spacy
!python -m spacy download en_core_web_sm
nlp=spacy.load("en_core_web_sm")
text='Book review: The Magic of the Lost Temple Authors: Sudha Murty
This is a wonderful story about a girl named Noonni who visits her
grandparents village'
doc=nlp(text)
noun_chunks = list(doc.noun_chunks)
print(noun_chunks)
print("Verb Phrases")
for token in doc:
    if token.pos == 'VERB':
        vp = " ".join([child.text for child in token.children if child.dep
in ["aux", "advmod"]]) + " " + token.text
        print(vp.strip())
print("Named Entities")
for ent in doc.ents:
    print(f"{ent.text} -> {ent.label}")
```

2)

```
import spacy

# Load the English language model
nlp = spacy.load("en_core_web_sm")

# Get user input (you can change this part as needed for testing)
sentence = input("Enter a sentence: ")

# Process the sentence with SpaCy
sentence = input("Enter a sentence")
doc = nlp(sentence)

# Extract noun phrases and verb phrases
noun_phrases = [chunk.text for chunk in doc.noun_chunks]
verb_phrases = [token.text for token in doc if token.dep_ == 'ROOT' or
token.dep_ == 'aux']

# Display results print("\nExtracted Noun Phrases:")
print(noun_phrases)
```



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```
print("Sentence: \n")
print(sentence)
print("\nExtracted Noun Phrases:")
print(noun_phrases)
print("\nExtracted Verb Phrases:")
print(verb_phrases)
```

Output

1)

```
Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
[Book review, The Magic, the Lost Temple Authors, Sudha Murty, This, a wonderful story, a girl, Noonni, who, her grandparents village]
Verb Phrases
Named Entities
Sudha Murty -> 380
Noonni -> 383
```

2)

```
Enter a sentenceThe foxes are jumping over the fence.
Sentence:

The foxes are jumping over the fence.

Extracted Noun Phrases:
['The foxes', 'the fence']

Extracted Verb Phrases:
['are', 'jumping']
```

Conclusion:

In conclusion, by leveraging both chunking and NLP methods, we can effectively identify and extract various syntactic structures from text. While chunking offers a quick and simple approach for phrase extraction, SpaCy provides a more robust and accurate solution by utilizing advanced machine learning models. Combining both techniques allows for more comprehensive text analysis, making it ideal for applications like information retrieval, sentiment analysis, and chatbot development.